

# Functional architecture for optical networks

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*ITU-T recommendation G.803 was originally developed to describe the functional and structural architectures of SDH transport networks. Many of the principles can be applied to transport networks in general including ATM and PDH. The existing model is extended to include wavelength division multiplexed optical networks and its application to passive optical networks, optical crossconnects and optical protection switching is described.*

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## 1. Introduction

Network operators are in the process of upgrading their transmission networks with the deployment of equipment conforming to the synchronous digital hierarchy (SDH). This provides many advantages over the existing plesiochronous digital hierarchy (PDH) including VC drop and insert capability thereby avoiding plesiochronous 'multiplexer mountains', direct visibility of bytes at all bit rates, simplified interworking, the introduction of the concept of optical midspan meet, provision of much more comprehensive overheads to provide end-to-end monitoring of the integrity and performance of a customer's circuit and automated network management. These benefits have been made possible through the development of a comprehensive set of standards by bodies such as the ITU-T, ETSI and ANSI. In addition to covering details of the hierarchy, equipment and interface specifications, these standards also cover architectural issues with SDH networks described in terms of a layered architecture, defined in ITU-T recommendation G.803 [1].

This paper examines how the principles of such a layered architecture can be extended to include research activity on transparent optical networks (TONs) which can be made transparent to data rate and format (within design limits) and are particularly suitable for bulk transport of high-bandwidth signals and services [2]. This type of network offers flexibility at the optical channel level and is based on wavelength division multiplexed (WDM) transmission and optical crossconnection, and it is shown how these can be described in a layered model. Section 2 discusses the concept of layered architectures and their representation based upon the existing recommendation, section 3 outlines how the concepts of a layered architecture can be applied to include some of the functions required in transparent optical networks, and section 4 describes some applications, including passive optical networks and protection switching.

## 2. Functional architecture of transport networks

During the development of the ITU-T recommendations on the SDH network node interface it became clear that there was a need for a supporting architectural recommendation [3, 4, 5]. This was considered necessary to provide a sound basis for the development of the associated equipment and network management recommendations. Architectural studies in the ITU-T have resulted in the production of recommendation G.803. Although G.803 describes the transport network architecture of SDH-based networks, much of the material is generic in nature and can thus be applied to transport networks independently of technology, e.g. ATM and PDH. Indeed work is now proceeding on a generic form of the recommendation (G.tna) [6].

The functional model used to describe transport networks is based on the concepts of layering, and, within each layer, partitioning in a manner which allows a high degree of recursion. A transport network can be decomposed into a number of independent layers, distinguished by the different types of signal carried, with a client/server relationship between adjacent layers. Each layer can be partitioned to reveal the structure within the network, either in terms of routing or topology, and in particular can be used to identify and describe significant boundaries. These boundaries may have administrative significance, such as those between network operators jointly providing end-to-end connectivity, or may be boundaries within one operator's domain for the purposes of defining management information flow and for apportioning performance objectives. The model uses a small number of architectural components which are defined either by the function they perform in terms of information processing, or by the relationships they describe between other architectural components. The points at which the inputs and outputs of processing

functions and transport entities are bound are the reference points of the transport network architecture. In this way it is possible to unambiguously describe the network both at a micro level (e.g. an individual network element or group of network elements in one location) or at a macro level (e.g. a complete layer network covering several administrative domains).

The diagrammatic conventions used for processing functions and reference points are shown in Fig 1 and are described below.

## 2.1 Topological components

These components define the topological relationships between sets of like reference points. Three topological components have been distinguished — the layer network, the sub-network and the link.

- A layer network is defined by the complete set of like access points which may be associated for the purpose of transferring information. Access point associations in a layer network may be made and broken by a layer management process thus changing its connectivity. A layer network is made up of sub-networks and links between them.

- A sub-network is defined by the complete set of like connection points which may be associated for the purpose of transferring characteristic information. The connection point associations in a sub-network may be made and broken by a layer management process thus changing its connectivity. Sub-networks are generally made up of smaller sub-networks and links between them.
- A link is defined by the sub-set of connection points in one sub-network which are associated with a subset of connection points on another sub-network for the purpose of transferring characteristic information between sub-networks. The set of connection point associations which define the link cannot be made or broken by the layer management process. The link describes the association between connection points contained in one network element with those in another.

## 2.2 Transport entities

These provide transparent information transfer between layer network reference points. There is no information change between input and output other than that resulting from degradations in the transfer process. Two basic entities are distinguished according to whether or not the information transferred is monitored for integrity. These are termed connections and trails. Connections are further

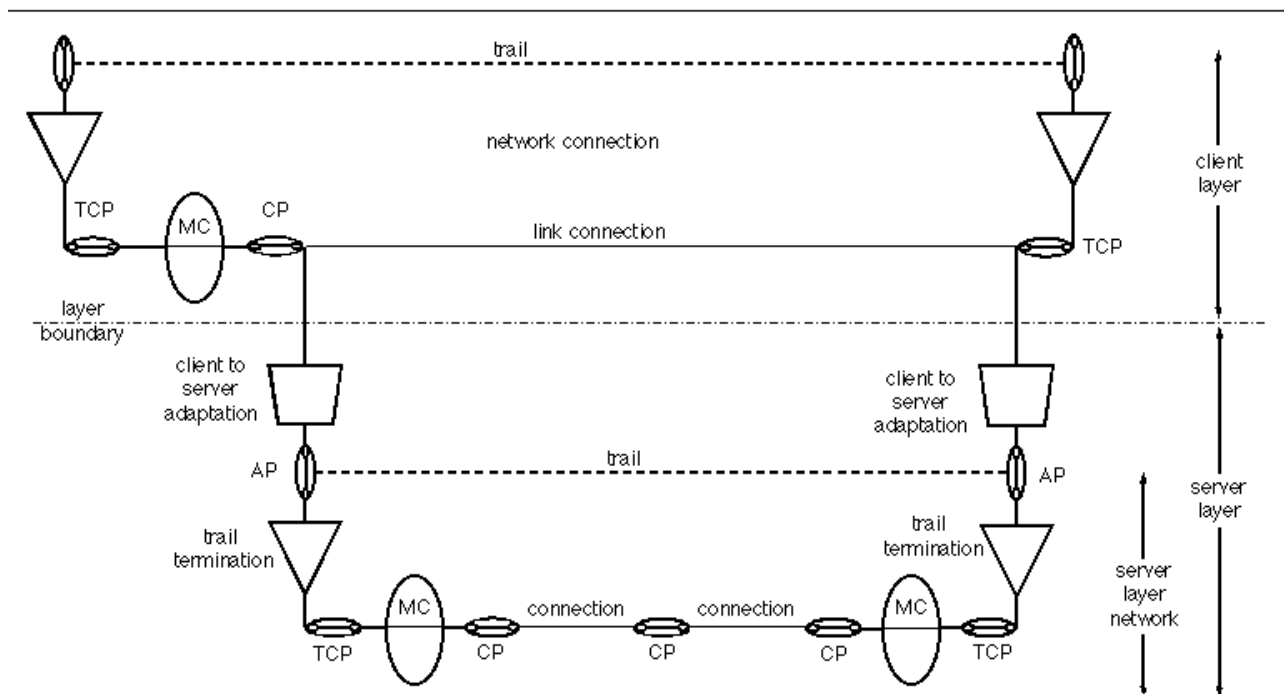


Fig 1 An example of a functional model and the use of some architectural components, (AP) access point, (TCP) termination connection point, (MC) matrix connection.

distinguished into network connections, sub-network connections and link connections according to the topological component to which they belong.

- A network connection is capable of transferring information transparently across a layer network. There is no validation of the transferred information.
- A sub-network connection is capable of transferring information transparently across a sub-network. Sub-network connections are, in general, made up of a concatenation of lower level sub-network connections and link connections.
- A link connection is capable of transferring information transparently across a link between two sub-networks. Link connections are provided by trails in the server layer network.
- A trail transfers validated characteristic information between access points, and it represents the association between access points together with any overhead added to validate the information transfer.

### 2.3 Transport processing functions

Two transport processing functions have been identified in describing the architecture of layered networks, these are the adaptation function and the trail termination function.

- The adaptation source function is the process whereby the characteristic information of the client layer network is adapted to a form suitable for transport by the server layer network. The complementary function of recovering the adapted information is the adaptation sink function. Typical processes which may occur in the adaptation function are coding, rate adaptation, aligning, justification and multiplexing.
- The trail termination function provides supervision of the information transfer in a trail. Typically this would be achieved by the addition of overhead information at a trail termination source function, this overhead information would be monitored at a trail termination sink function.

### 2.4 Reference points

In a layer network, reference points are located between the output of one transport processing function or transport entity and the input of another.

### 2.5 The layering concept

The layering model depicted in Fig 2 contains three classes of layer networks.

- Circuit layer networks are oriented towards public switched services such as the public switched telephone network (PSTN), integrated services digital network (ISDN), broadband integrated services digital network virtual channels (B-ISDN VCs) and X.25, etc. Circuit layer networks are independent of path layer networks.
- Path layer networks are oriented towards the provision of the links in the circuit layer network such as 2 Mbits, 8 Mbit/s, 34 Mbit/s and 140 Mbit/s for PDH, higher-order (VC-4) and lower-order (VC-11 and VC-12) SDH VCs, B-ISDN virtual paths, etc. Path layer networks are independent of transmission media layer networks.
- Transmission media layer networks are normally oriented towards point-to-point transport across line or radio transmission systems.

Trails in each of these types of layer networks are called circuits, paths, and sections respectively. Transmission media layer networks are divided into section layer networks and physical media layer networks. Section layer networks are concerned with the transfer of information between two nodes in path layer networks, whilst physical media layer networks are concerned with the actual optical fibre, metallic pair or radio frequency bearer which supports a section layer network. In the case of SDH, there are two section layer networks — a multiplex section layer and a regenerator section layer network. The multiplex section layer network is concerned with the transfer of information between nodes which route or terminate paths, while the regenerator section is concerned with the transfer of information between individual regenerators and between regenerators and nodes which route or terminate paths.

### 2.6 Extending the layering concept

The transport network layered model shown in Fig 2 represents a conventional SDH network, but there is no reason why this model could not be extended upwards to include application layer networks supported by the circuit layer network and downwards to include optical sections, fibre, cable and duct layer networks.

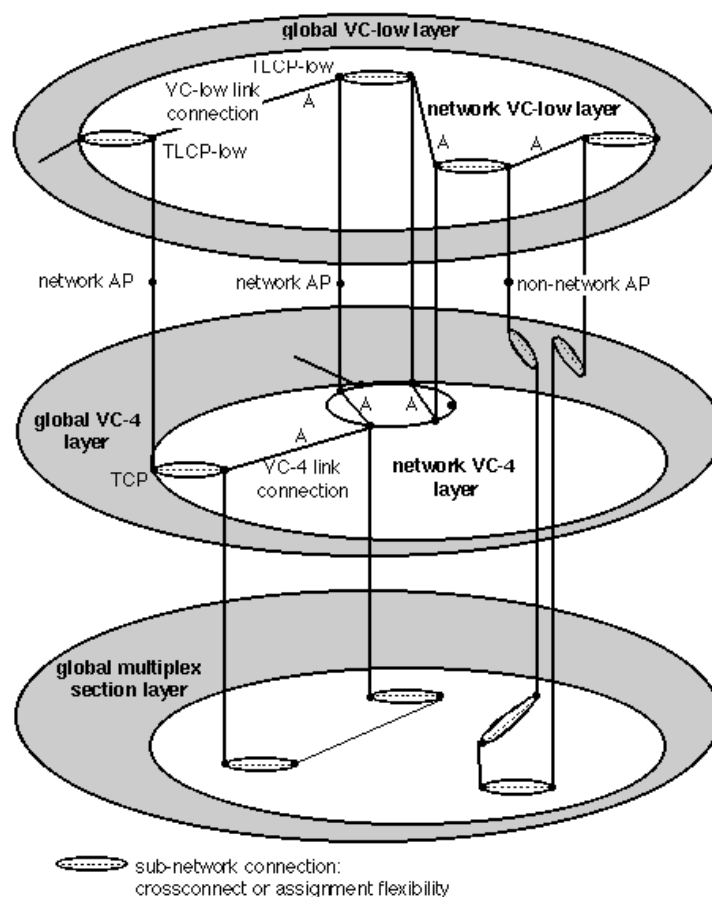


Fig 2 Example of SDH network layers illustrating a lower order path layer (VC-low), a higher order path layer (VC-4) and a multiplex section layer.

A number of organisations are currently involved in research activity on TONs which employ WDM transmission and optical crossconnection, and as a result there is considerable interest in modelling these networks. In order to ensure that these new networks can be managed in a way which is consistent with SDH networks, it is important that the same functional model is used.

### 3. Layered model for optical networks

In developing a model to support transparent optical networks it is not necessary to describe what occurs in the higher layers which may be based on PDH, SDH, ATM or some other format, only the transmission media-dependent layers need be considered. Conventionally this layer is further subdivided into section layers and a physical media layer. While this classification is neither absolute nor essential, it provides a means of identifying the purpose of each layer and alternative proposals exist in the research literature for the introduction of an optical path layer [7, 8]. Within the context of G.803, path layers are defined as being independent of transmission media layer networks (i.e. a path can be supported over any transmission media fibre, coaxial cable, radio, satellite, etc) and as such the definition of an optical path layer or reference to existing SDH

path layers as electrical layers is inappropriate. However, there is no intrinsic difference between a media-dependent path layer and a section layer which is networked — only in terminology. This paper follows convention although the names of the transmission media layers and the functions contained within these layers are defined so as to avoid any confusion with SDH specific terms. The number of optical layers that are required for an optical network depends upon whether the network utilises single or multiple optical channels.

#### 3.1 Single channel systems

Networks based on single optical-channel working can be described by means of a fibre layer. The characteristic information of the fibre layer is the optical signal, hence adaptation from, say, the SDH regenerator section to the fibre layer is the adaptation from a frame structure into an optical power level, pulse shape and wavelength (or frequency) and its mapping into a fibre. The termination function is the generation and measurement of the optical power level.

#### 3.2 Multiple channel systems

For multiple optical-channel working the adaptation function in the fibre layer needs to be expanded to introduce

a new layer that allows for multiple STM-Ns (or other kinds of section) on one fibre. This is achieved by introducing an optical channel section layer (OCS-layer) between the upper section layers (i.e. the regenerator and multiplexer section layers) and the fibre layer. The adaptation from the upper section layer into the OCS-layer is from say an SDH STM-N frame into an optical channel of given bandwidth and the associated overhead is the average optical power level. A further overhead may be added to the optical channel by means of a low frequency modulation. The source trail termination function (TTF) is the generation of the optical power level or the modulation signal, while the sink (TTF) is the measurement of the optical power level either directly or from the demodulated signal. If the STM-N frame cannot be recovered at the sink adaptation, that is an adaptation failure. It should not be assumed that adaptation from the OCS-layer to the upper section layer always involves optical-to-electrical conversion. In this example, the characteristic information of the upper section layer is the STM-N frame structure which is a logical information pattern. External to the equipment this is represented by the

arrangement of photons whilst internally it may be represented by any means the manufacturer may choose to use.

An OCS-layer contains single optical channels and there is no explicit reference to wavelength in this layer. Wavelength is the means for distinguishing between different optical channels within the same fibre, but for any one channel, wavelength is not an attribute outside the context of a fibre. In traversing an OCS-layer an optical channel may have different wavelengths on different fibre links as a consequence of switching or wavelength conversion. This is analogous to 64 kbit/s channels which do not have timeslots except when they are inside 2 Mbit/s frames and normally have different timeslots on different sides of a 64 kbit/s switch.

The adaptation from the OCS-layer to the fibre layer involves the assignment of absolute wavelength (or frequency) and multiplexing of individual optical channels. Figure 3 illustrates how a number of optical layers connect.

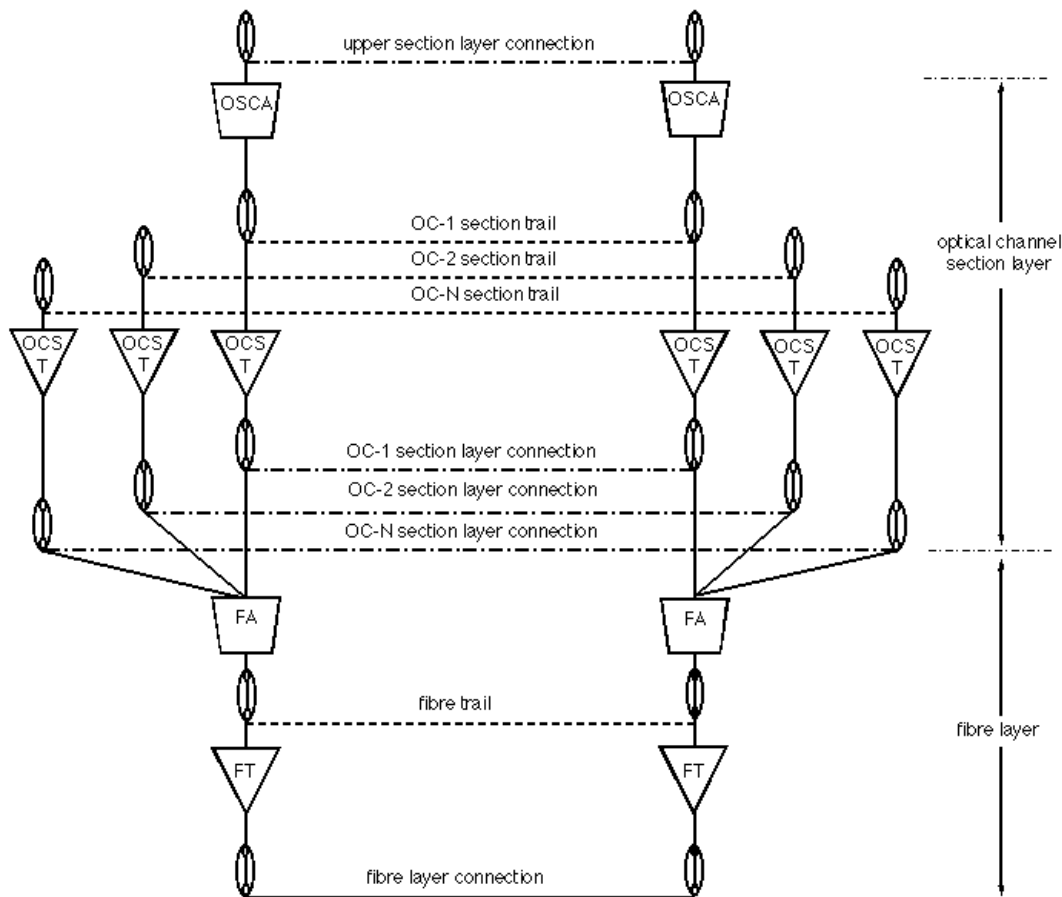


Fig 3 Application of functional architecture to case of optical transmission layers — (OSCA) optical channel section adaptation, (OCST) optical channel section termination, (FA) fibre adaptation, (FT) fibre termination.

The overhead in the fibre layer would most probably be an optical pilot (which could be modulated to provide extra information-overhead capacity) and loss of the pilot would indicate failure of the fibre layer.

Further layers can be added to this model (or existing layers expanded) depending upon the levels of networking required. A group of optical channels can be divided into smaller logical groups which are networked as entities in their own right. As an example hierarchical routing could be based on telecommunications windows, i.e. 1300 nm or 1500 nm, or on 10 nm portions of each window, and so on. This leads naturally to the concept of channels, groups and supergroups similar to those described in existing FDM systems [9]. These additional layers are not, however, considered in the remainder of this paper.

## 4. Applications

### 4.1 Representing access networks

In the previous section, multiplexing of optical channels was shown at a single location and this can be extended to include passive optical networks (PONs) for local access.

As an example consider a TDM signal broadcast to all terminals from the exchange on a single optical channel, with individual customers accessing the appropriate timeslots and a second optical channel used in the upstream direction. The transmission media layers of such a network are shown in Fig 4 and they have the following features:

- the PON can be represented in the downstream (upstream) as a single 1-to-N splitter (N-to-1 combiner) in the fibre layer or as a number of smaller splitters (combiners) interconnected by links,
- the fibre layer reference points are bidirectional while those of the optical-channel section layer are unidirectional — this indicates that the optical channel in the downstream direction can support a different bit rate and/or frame format from the upstream channel,
- point-to-multipoint trails are possible in the downstream direction in both the optical channel section layer and fibre layer — if a path layer exists above the optical layers, its associated trails may be either point-to-point or point-to-multipoint,

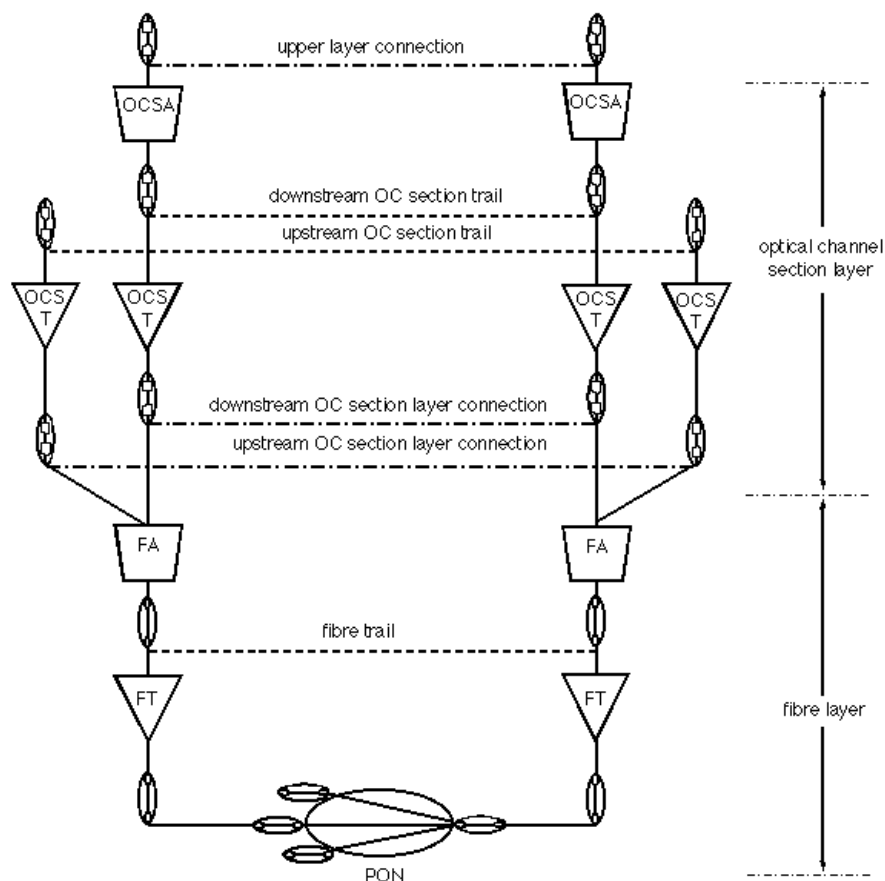


Fig 4 Example of a passive optical network architecture.

- additional optical channels can be provided in both directions — these new channels may support new services and frame or packet formats.

#### 4.2 Representation of optical amplifiers

Modelling optical amplifiers can be approached from the basic functional point of view. A server layer contracts (implicitly or explicitly) with a client layer to transport a client layer-link connection to a certain level of performance. All server layer components are liable to cause degradation of the characteristic information passing through, so it is a responsibility of the server layer either to use components whose inherent degradation is acceptably low, or alternatively, to employ active methods for combating the degradation. Examples of the latter are re-transmission of lost or errored frames, or forward error-correction codes, which allow layers acting as clients to use poorer quality server layers below them. Ensuring integrity of information is a trail termination function.

In an optical fibre, losses cause gradual degradation of the characteristic information (the total optical signal which

a fibre can pass). As the signal diminishes, the signal-to-noise ratio reduces, increasing the risk of errors in a digital signal or reducing the quality of an analogue signal. The trail termination sink function will monitor the fibre integrity (e.g. by measuring the level of an optical pilot) to determine whether the fibre loss is excessive, even if it cannot take any corrective action. If the fibre is not too long or the loss too high, this degradation will be within acceptable limits. If the fibre is too long, and nothing else were done, the degradation would be excessive and the trail termination sink function would report fibre failure. If, therefore, an optical amplifier is added, it is in order to counter degradation and preserve the integrity of the characteristic information. It is therefore a termination function.

At each amplifier, the fibre layer is terminated, so that characteristic information (partly dependent on signal-to-noise ratio) is not simply passed through unchanged. Certainly the characteristic information of the client layers is not modified. Corrective action is taken (amplification), to preserve the signal-to-noise ratio from excessive degradation, and the characteristic information is sent out

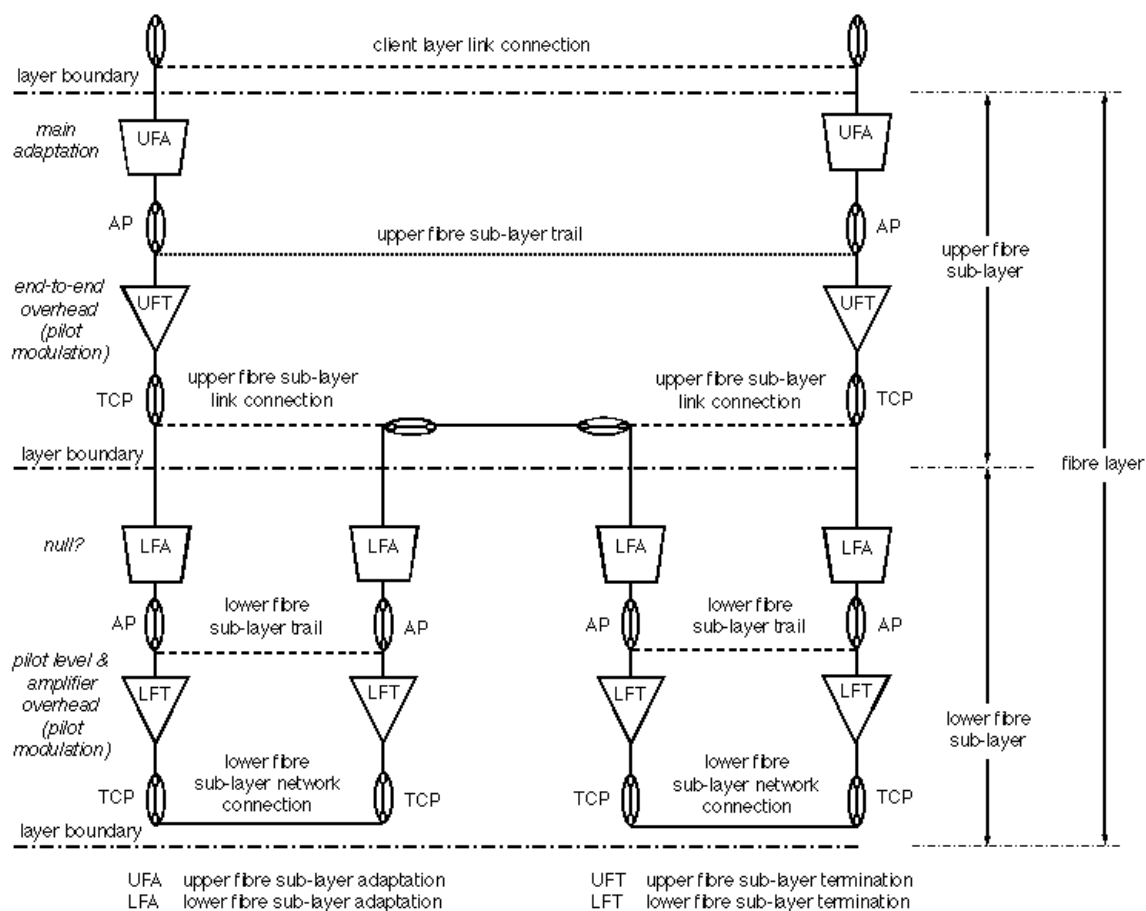


Fig 5 Introduction of sub-layer for optical amplifiers.

with a resemblance of what it was when it left the terminal equipment at the sending end of the fibre link connection. There is no requirement for the outgoing overhead to be restored to any absolute norm; if the incoming pilot level is lower than it should be after allowing for expected fibre loss, then the outgoing pilot level may also be sent at a lower level. This can be considered as the overhead passing on information about the quality of the signal, i.e. there is some degradation, but much less than if the amplifier had failed.

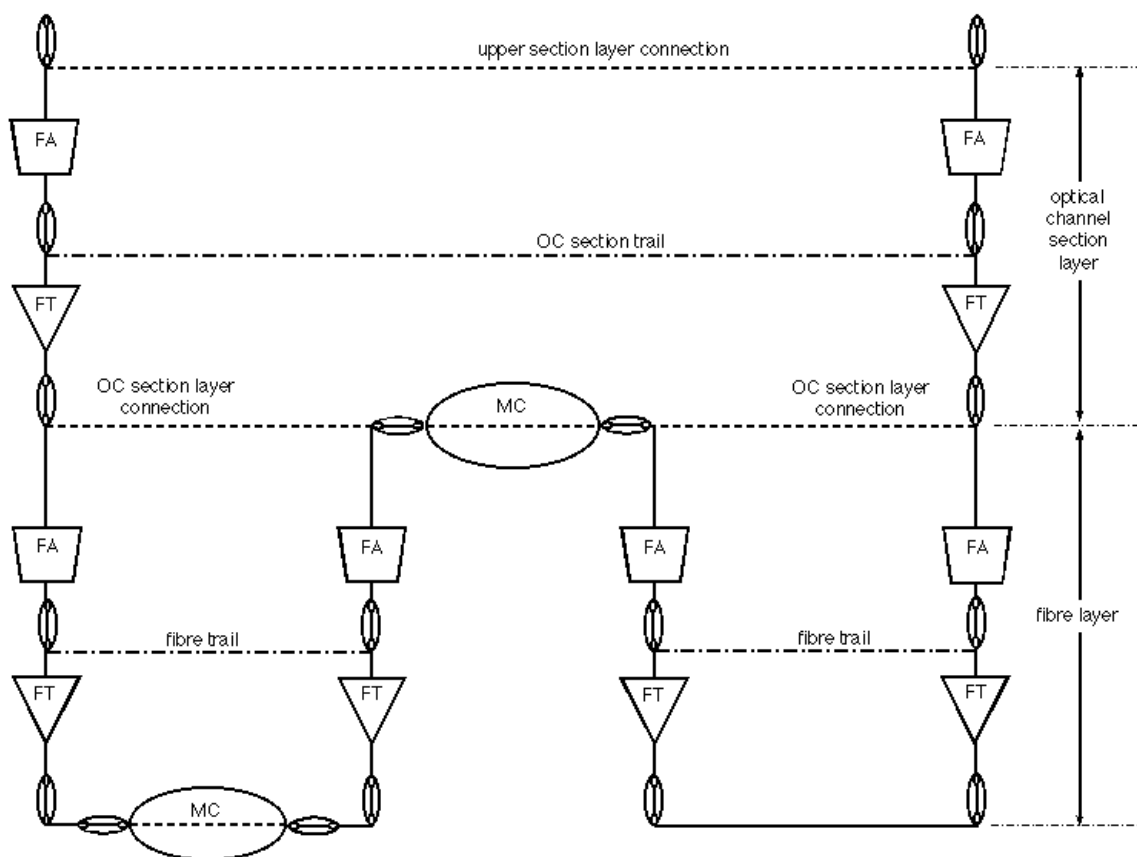
The amplifier is carrying out this termination function quite independently of whether it deliberately monitors power level and generates any kind of management reports. If it does not do these things, then it is simply that the management information is not passed on anywhere (and not deliberately collected). If monitoring is done then this can be explicitly shown as a management interface on the termination, with adaptations and terminations as required to access a management message channel (either in the same fibre or via some external route).

In the amplifier, it is not allowed to show simply two back-to-back terminations; adaptation and termination

always go in pairs, taking a signal from one layer to another. However, it is not necessary to show the amplifier de-adapting up to the fibre's client layer; the amplifier function can be shown as a termination in the lower fibre sub-layer. The lower fibre sub-layer adaptation might therefore be null. Thus the amplifier connects two sub-layer link connections together as shown in Fig 5.

An optical pilot used for monitoring the fibre may well be modulated to carry management information about the fibre layer. Although, in principle, a sub-layer for each bit and byte of the modulation can be shown, in practice it may well be sufficient to put in the upper fibre sub-layer those parts which the amplifiers pass transparently, and in the lower sub-layer those parts which they do or could modify.

When there is no monitoring of the characteristic information within an amplifier it is necessary to consider whether a sub-layer is necessary at all for instantiation. It may not be applicable where the amplifier is being used as a booster at a source or as a pre-amplifier at a sink and is part of an equipment's internals, but may very well be necessary for a stand-alone amplifier in a fibre link outside an exchange building.





Although this section has described the modelling of amplifiers, the model is true of any active mechanism in the lower fibre sub-layer that is used to preserve the integrity of the characteristic information.

#### 4.3 Representation of switching

Switching can occur at two levels in an optical network, namely the complete contents of a fibre (i.e. all optical channels on a fibre are switched as a single bundle) or as individual channels. The first would therefore occur in the fibre layer whilst the second would occur in the optical channel section layer. This is illustrated in Fig 6 where only one optical channel is shown for clarity but others can be added at the fibre adaptation level. A switch is represented by means of a matrix connection (MC) or a sub-network. In the fibre layer the wavelength of the optical channel would be the same before and after switching, while in the section layer the output need not be the same as that of the input.

#### 4.4 Optical protection strategies

A failure or degradation of quality within a transport network can be managed either through protection which makes use of pre-assigned capacity between nodes or by restoration which utilises any available capacity between nodes. Switches in both the optical layers can be used for either purpose. The simplest protection architectures are 1:1 (see Fig 7) and 1+1 and offer complete protection against fibre failure when applied with a physically diverse protection fibre in the fibre layer.

The example of 1:1 protection in Fig 7 shows where trail signal failure (in this case loss of optical power) in the fibre termination is used to reconfigure the protection switches. As the operation is dual ended, a fibre automatic protection switching (FAPS) channel is required between the switches in the fibre layer overhead (e.g. the optical pilot mentioned above). In a 1+1 architecture the switch at the transmitting end is replaced with a permanent bridge such as a 1 x 2 fibre splitter and a FAPS is not required.

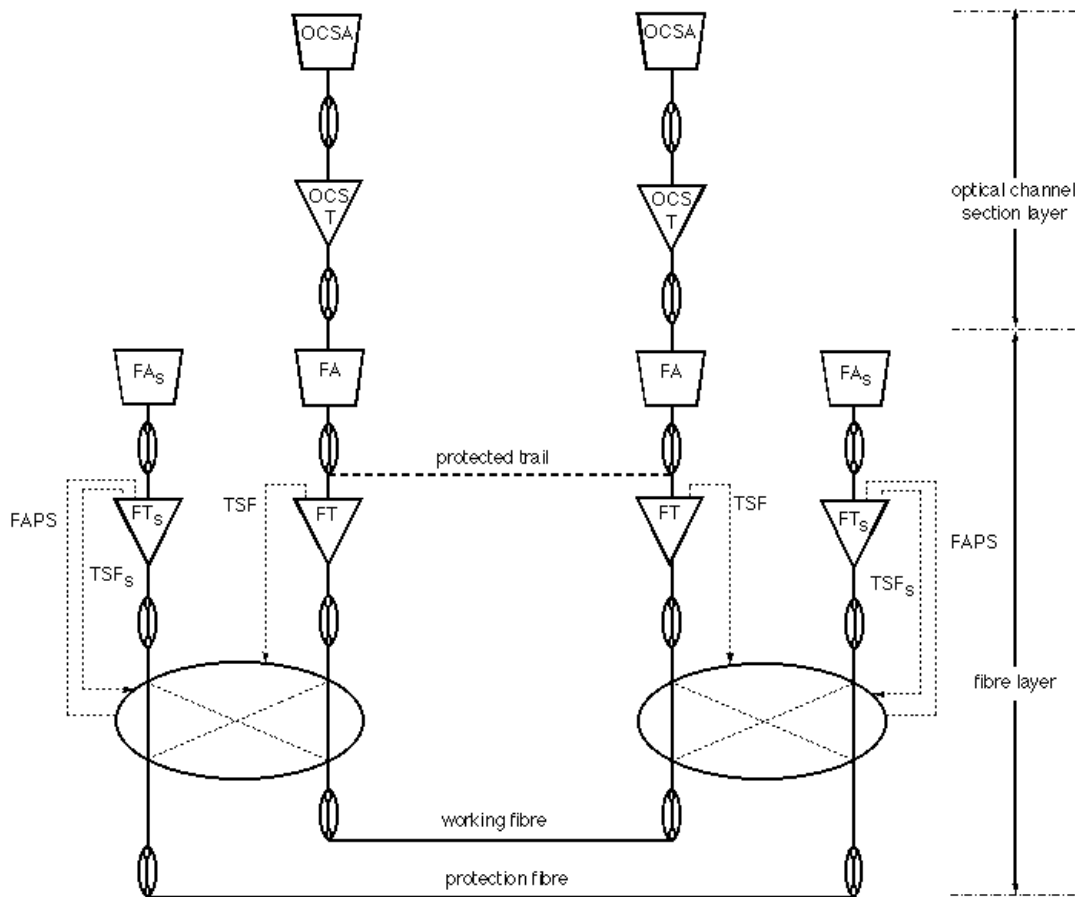


Fig 7 A possible 1:1 optical protection switching architecture. Switching occurs in response to trail signal failure (TSF) in the fibre layer and the FAPS protocol. The protection fibre is supervised by means of a test signal on the fibre (FT<sub>S</sub>) and in the event of a TSF<sub>S</sub> switching to the protection fibre would be prevented.

#### 4.5 Optical management channels

In an SDH ring, two nodes (one main, one stand-by) known as gateway network elements (GNEs) allow remote management centres to communicate with the rings. Messages between the other ring nodes and the GNEs are provided by the data communication channel (bytes D1 — D12 in the SDH section overhead). In a similar manner management information for the fibre layer including FAPS can be transferred by means of the fibre layer overhead between optical elements such as crossconnects and amplifiers.

#### 5. Conclusions

This paper has discussed how the existing G.803 model for SDH networks could be extended to include new network architectures based on wavelength division multiplexing, amplification and switching of both single- and multiple-channel systems. These techniques have been outlined in a number of examples and, while it can be anticipated that a wide variety of networks will be developed during the process of research into optical networks, the principles outlined here can be applied as they are generic and independent of technological implementation. The transparency of these networks to bit rate and format means that, for the purpose of network management, non-intrusive monitoring and/or external supervisory channels are required rather than intra-channel processing. The use of such a model therefore provides a strong foundation for the development of management information models and for definition of the interaction between existing and future management platform

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Andrew Nunn joined BT in 1963 as an apprentice. After a two year training period he worked on all aspects of transmission maintenance. In 1980 he joined Telconsult and spent eighteen months in Libya where he supervised the installation and testing of transmission equipment and external plant.

He returned in 1982 to be the manager responsible for transmission, radio and cable TV maintenance in Colchester. In 1989 he joined BT Laboratories to work in the network standards area.

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Alan Jackson graduated from Birmingham University with a degree in electronic engineering in 1971 and joined BT to work on procurement specifications and prototype testing of the last generation of FDM line system (60 MHz), and then on FDM multiplexers, and analogue and optical fibre field equipment.

More recently he worked on transmission network element management interfaces. He now provides guidance on, and promotes, good network architecture, particularly for transport networks, and facilitates the generation of commercial direction for BT's

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Goff Hill joined BT in 1961 as an apprentice. He went on to win a BT scholarship to attend the University of Newcastle upon Tyne and in 1969 gained an honours degree in electrical engineering. Since then he has worked at BT Laboratories on local networks, visual system and, for the last twelve years, on optical communications. Currently he leads the future network infrastructure group within the networks research unit. The group has developed concepts of optical transport networks and is showing how they can be applied in BT's network. One of his major tasks over the past three years has been to lead the RACE II 'Multi-Wavelength Transport Networks' project which has established a high world-wide profile through demonstrating optical network principles. The group is now directing much of its effort towards exploring how emerging technologies will impact on BT's future network.